Portfolio Case Study — Fedor Gorbunov
AI Agent Operator Cockpit for Long-Running Codex Sessions

Executive Summary
I designed and built a private operator cockpit for long-running AI agent sessions. The goal was not to create another chatbot UI, but to build a reliable control surface where the operator can see what the agent is doing, where prompts are going, which session is active, what needs attention, and which artifacts were produced.

Business Problem
AI agents are useful only when their work is observable and controllable. Long-running sessions often fail because the operator cannot clearly see runtime state, session context, task progress, documentation, recovery paths, or the boundary between the agent’s output and the real system state.

What I Built
The cockpit combines dialog control, runtime status, planning, documentation, board-based reasoning, translation support, bookmarks, terminal/log observation, and session artifacts in one workspace. The dialog is the primary control surface, while terminal/log views act as observer surfaces rather than hidden execution channels.

Architecture Highlights
- Dialog-first control model for AI-agent sessions.
- Runtime registry for live/dead sessions, bound terminals, and recovery actions.
- Session-bound artifacts: plans, notes, docs, board snapshots, and logs.
- Fail-closed status model: UI state should reflect runtime truth, not optimistic assumptions.
- Operator workflow support: active work package, documentation, translation, and visual reasoning stay near the live session.

My Role
Founder, AI Systems Architect, Full-Stack Builder, and Operator. I defined the product model, requirements, architecture, acceptance rules, UI flow, runtime truth model, and AI-assisted delivery workflow.

Result
The system creates a more reliable AI-assisted engineering workflow: the operator can supervise agent work, track state, recover sessions, inspect artifacts, and keep documentation connected to execution.

Not Disclosed
Private account data, tokens, internal repository paths, full repository structure, and private runtime/session internals.